

10 Hard Fix the Error Questions

Question 1: Binary Search Bug

```
while(l<r){... l=mid; r=mid;}
```

Task: Identify and fix the bug(s).

Question 2: Factorial Base Case

```
if(n==0) return 0;
```

Task: Identify and fix the bug(s).

Question 3: Vector Index

```
vector<int> v={10}; cout<<v[1];
```

Task: Identify and fix the bug(s).

Question 4: Missing Header

```
sort(a, a+5);
```

Task: Identify and fix the bug(s).

Question 5: Infinite Loop

```
while(i<5){ cout<<i; }
```

Task: Identify and fix the bug(s).

Question 6: Assignment in If

```
if(a=5)
```

Task: Identify and fix the bug(s).

Question 7: Array Bounds

```
for(i=0;i<=5;i++)
```

Task: Identify and fix the bug(s).

Question 8: Uninitialized Sum

```
int sum;
```

Task: Identify and fix the bug(s).

Question 9: Pattern Bug

```
for(...); cout<<'*';
```

Task: Identify and fix the bug(s).

Question 10: Division by Zero

```
cout<<x/0;
```

Task: Identify and fix the bug(s).

Answer Key

Q1: Use while($l \leq r$), $l = \text{mid} + 1$, $r = \text{mid} - 1$

Q2: return 1

Q3: Use $v[0]$

Q4: Add `#include <algorithm>`

Q5: Add `i++`;

Q6: Use `==`

Q7: Use `i < 5`

Q8: Initialize `sum = 0`

Q9: Remove `;` and print inside loop

Q10: Check denominator